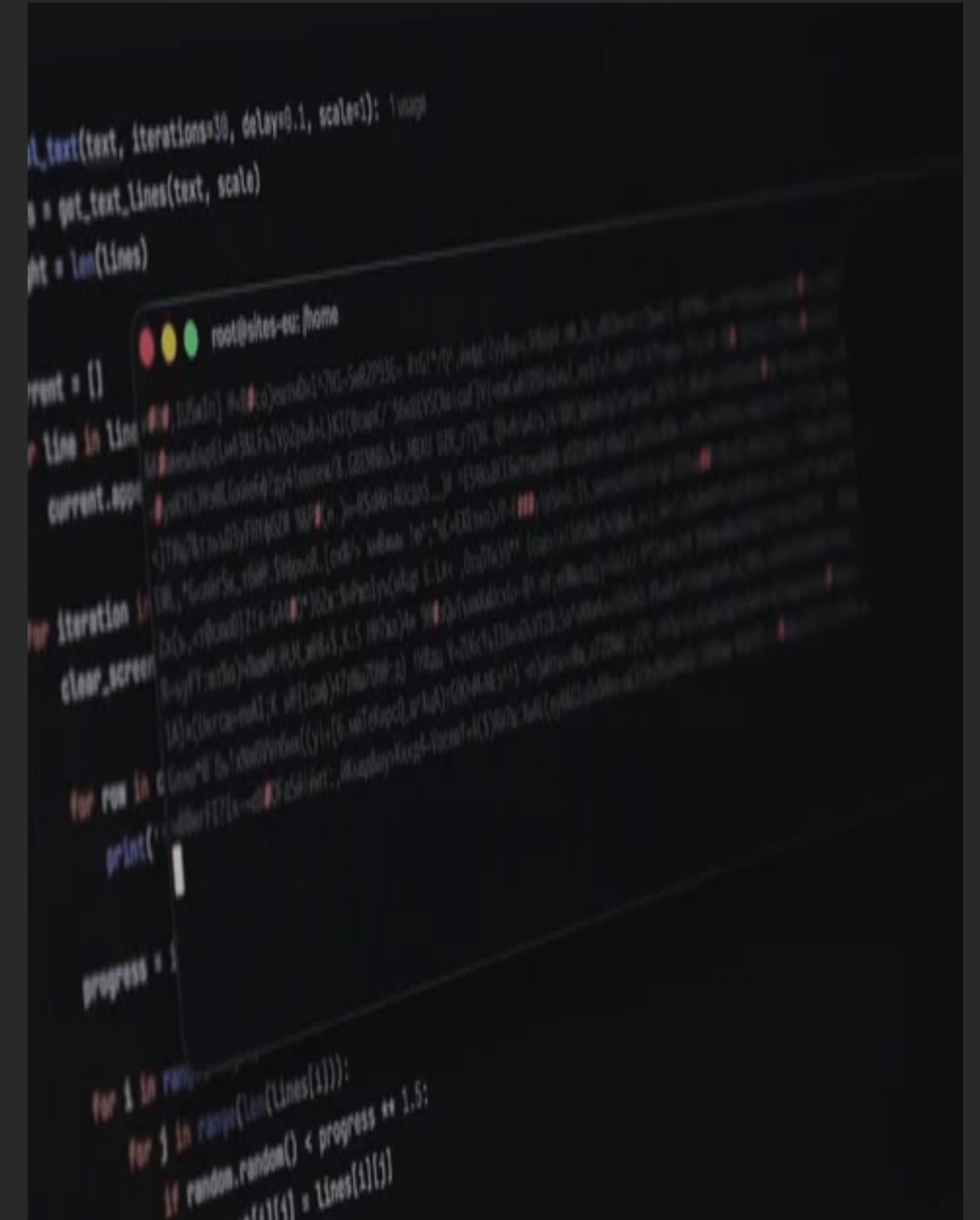

DEBUGGING MODERN FULL-STACK APPLICATIONS

Think Like a Software Engineer

Week 8 — Full-Stack Development Course

Instructor: Dilovan Matini

Provided by CAPA & GIZ



Today's Topics

What we will cover in this session

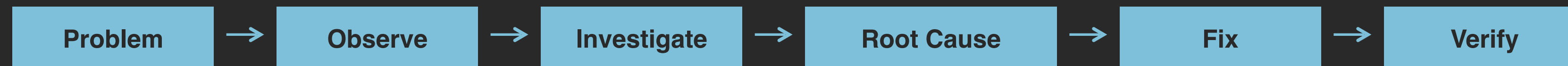
- What is Debugging?
- Types of Software Bugs
- Debugging Workflow
- Backend & Frontend Debugging
- Browser DevTools & React Tools
- HTTP Status Codes
- Reading Error Messages
- Debugging Examples
- Professional Debugging Tips
- Classroom Challenge — Bug Hunt

What is Debugging?

Definition

Debugging is the process of finding, understanding, and fixing problems in software.

Professional Workflow



Never guess. Always investigate.



Types of Bugs

Syntax Errors

Missing brackets, missing commas, invalid JavaScript

Runtime Errors

Undefined variables, missing imports, null values

Logic Errors

Wrong calculations, wrong conditions, incorrect output

API Errors

Wrong URL, wrong HTTP method, wrong response

UI Errors

Button not working, data not displayed, styling problems

Debugging Workflow

Read the Error



Find the File



Locate the Line



Understand



Fix the Code



Test Again



Remember: **Reading the error message usually gives you the answer.**



Backend Debugging

Tools

- VS Code
- Terminal
- `console.log()`
- REST Client

Common Problems

- Syntax errors
- Undefined variables
- Wrong routes
- Missing middleware
- Server not running



Frontend Debugging

Tools

- Browser Console
- Network Tab
- Elements Tab
- Sources Tab
- React Developer Tools

Common Problems

- Wrong imports
- Wrong props
- Wrong state
- Fetch errors
- Rendering issues

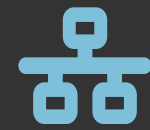


Browser DevTools



Console

Display logs, warnings, and errors from your JavaScript code



Network

Inspect all API requests and responses in real-time



Elements

Inspect and modify HTML structure and CSS styles



Sources

Use breakpoints to pause and inspect code execution

Press **F12** (Windows) or **Cmd+Option+I** (Mac) to open DevTools

React Developer Tools

React Developer Tools helps you inspect:



Components

See the component tree and hierarchy



Props

Check what data is passed to each component



State

Monitor state changes in real-time



Hooks

Debug useState, useEffect, and custom hooks

Instead of **guessing** what React is doing, **inspect it directly**.

HTTP Status Codes

Code	Meaning	Category
200	OK	Success
201	Created	Success
400	Bad Request	Client Error
401	Unauthorized	Client Error
403	Forbidden	Client Error
404	Not Found	Client Error
500	Internal Server Error	Server Error

Understanding status codes helps **identify backend problems quickly**.

- 2xx = Success
- 4xx = Client Error
- 5xx = Server Error

Reading Error Messages

Example

ReferenceError

books is not defined

app.js:15

Always Identify

- **Error Type** — ReferenceError
- **Error Message** — books is not defined
- **File Name** — app.js
- **Line Number** — 15



Do not ignore the stack trace.

Backend Debugging Example

Code

```
app.get("/books", (req, res) => {  
  res.json(bookList);  
});
```

Question

Why does this code fail?

Hint

There is no variable called **bookList**.

Answer

The variable **bookList** is not defined.

You need to define it first:

```
const bookList = [];
```

Frontend Debugging Example

The Component

```
<BookCard title={book.name} />
```

Question

Why isn't the title displayed?

The Object

```
{  
  title: "Clean Code"  
}
```

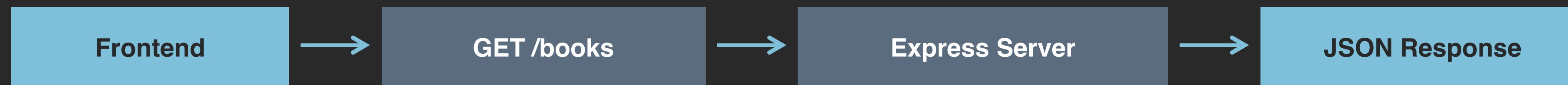
Answer

The code uses **book.name**
but the property is called **title**.

Fix: Use **book.title** instead.

API Debugging

Request Flow



Debug Using

- REST Client
- Network Tab

Ask Yourself

- Is the URL correct?
- Is the HTTP method correct?
- What is the response?

Professional Debugging Tips

- ✓ Read the error carefully
 - ✓ Fix one problem at a time
 - ✓ Test after every change
 - ✓ Use `console.log()`
 - ✓ Use DevTools
- ✓ Don't panic
 - ✓ Don't copy random code from the Internet
 - ✓ Understand the root cause

Golden Rule: Understanding the problem is more important than fixing it quickly.

Classroom Challenge — Bug Hunt

Mission

Each team will receive a Book Store application containing several bugs.

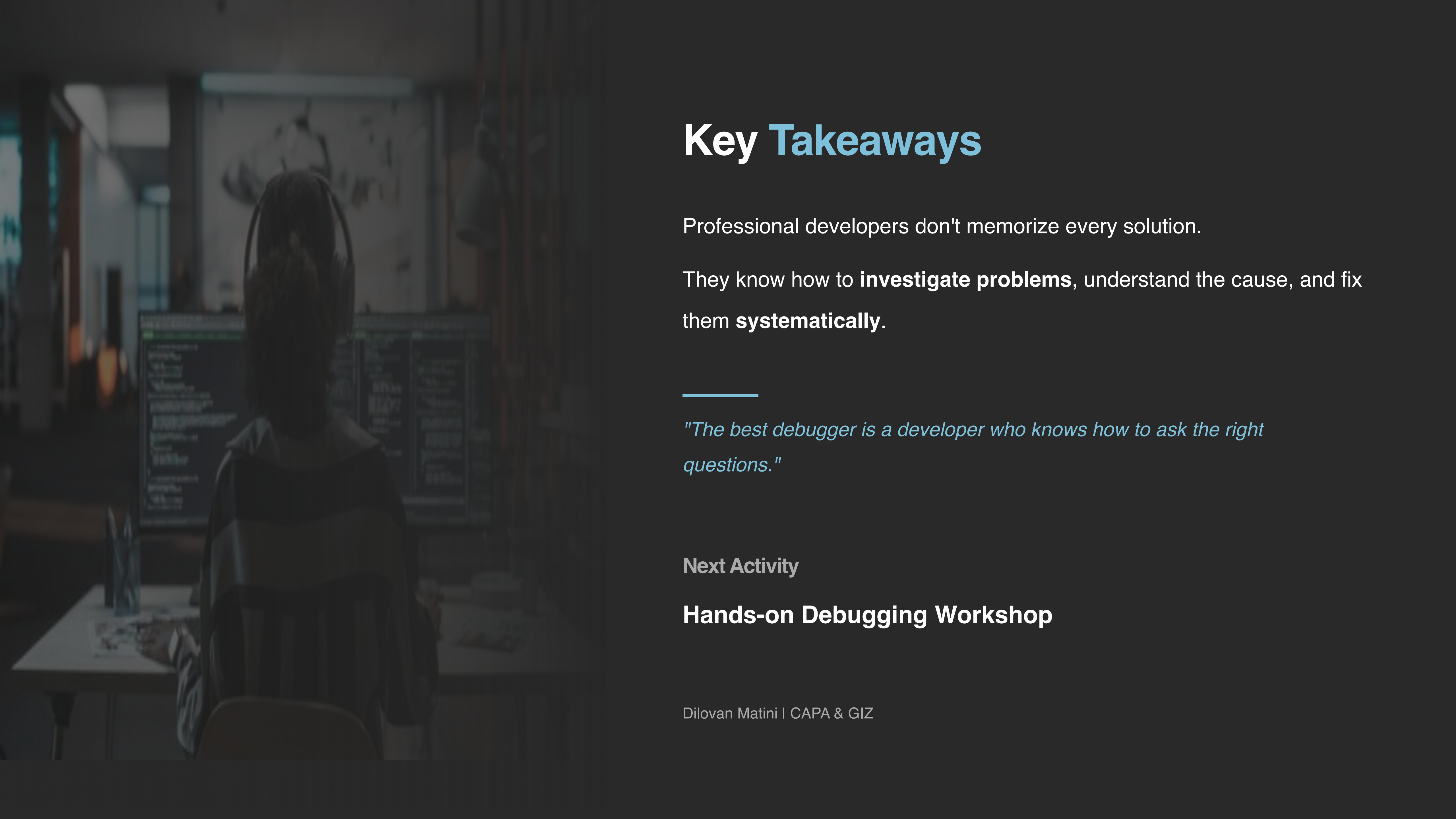
Your Tasks

- 1. Identify each bug
- 2. Explain the cause
- 3. Fix the problem
- 4. Test the application
- 5. Demonstrate the solution

Tools Allowed

- VS Code
- Browser DevTools
- React Developer Tools
- REST Client



A person with long dark hair, wearing large headphones, is seen from behind, sitting at a desk in a dimly lit office. They are looking at two computer monitors displaying code. The background is dark and out of focus, showing office furniture and lights.

Key Takeaways

Professional developers don't memorize every solution.

They know how to **investigate problems**, understand the cause, and fix them **systematically**.

"The best debugger is a developer who knows how to ask the right questions."

Next Activity

Hands-on Debugging Workshop

Dilovan Matini | CAPA & GIZ